Lecture notes on

Probability

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Contents

1 Probability axioms & conditional probability

Definition. The *sample space* S of an experiment is the set of all possible outcomes. An *event* A is a subset $A \subseteq S$ of the sample space. A *probability* is assigning some value in $[0,1] \subset \mathbb{R}$ to an event. A *probability space* (S,P) consists of a sample space S and a probability function P mapping each $A \subseteq S$ to a number $P(A) \in [0,1]$, satisfying the following properties, known as the *Kolmogorov axioms*:

1. $P(\varnothing) = 0$, $P(S) = 1$
2. *Countable additivity* If $A_1, A_2, \dots \subseteq S$ are disjoint events ($A_i \cap A_j = \varnothing$ for all $i \neq j$) then

$$P(\bigcup_{j=1}^{\infty} A_j) = \sum_{j=1}^{\infty} P(A_j)$$

This is all we need to do probability. A weaker form of countable additivity, called *finite additivity*, follows: for any finitely many disjoint sets,

$$P(\bigcup_{j=1}^{n} A_j) = \sum_{j=1}^{n} P(A_j)$$

Theorem. Suppose (S,P) is a probability space. Then, for any events $A_1, \cdots, A_n \subseteq S$,

1. (*Complement rule*) $P(A_1^c) = 1 - P(A_1)$ where $A_1^c = S \backslash A_1$.
2. (*Monotonicity*) If $A_1 \subseteq A_2$ then $P(A_1) \leq P(A_2)$.
3. (*Inclusion-exclusion*)

$$P(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) + \dots + (-1)^{n+1} P(A_1 \cap \dots \cap A_n)$$

For a finite sample space, the *uniform probability function* assigns equal weight to all outcomes in the sample space.

Let A and B be events with $P(B) > 0$. Then, define the *conditional probability*

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Theorem. (*Bayes' rule*) For any events A and B with $P(A) > 0$ and $P(B) > 0$,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Theorem. (*Law of total probability*) Suppose $B_1, B_2, \cdots$ are a partition of the sample space. Then for any event A,

$$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i) = \sum_{i=1}^{\infty} P(A|B_i)P(B_i)$$

Example. A disease affects 1% of the population. A test for the disease is advertised to be 95% effective, meaning that if a person is sick, there's a 95% chance they test positive. If a random person is tested and tests positive, what's the probability that they are sick?

Denote having the disease as D and testing positive as T, so $P(D) = 0.01$ and $P(T|D) = P(T^c|D^C) = 0.95$. We wish to solve for $P(D|T)$. By Bayes' rule,

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)}$$

By the law of total probability, since D, D^c form a partition, $P(T) = P(T|D)P(D)+P(T|D^c)P(D^c) = P(T|D)P(D) + (1 - P(T^c|D^c))(1 - P(D))$. So

$$P(D|T) = \frac{P(T|D)P(D)}{P(T|D)P(D) + (1 - P(T^c|D^c))(1 - P(D))} = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + (1 - 0.95)(1 - 0.01)} \approx 0.16$$

meaning that there's only a 16% chance that the random person who tested positive is actually sick.

Definition. Two events A and B are *independent* if $P(A \cap B) = P(A)P(B)$.

Prop. If $P(B) > 0$, A, B independent $\iff$ $P(A|B) = P(A)$ (and similarly for $P(A)$).

Definition. Events A and B are *conditionally independent* given a third event E (with $P(E) > 0$ if

$$P(A \cap B|E) = P(A|E)P(B|E)$$

Note that conditional independence and independence are independent.

Note. From this point onwards, we write $P(A, B) \equiv P(A \cap B)$.

2 Discrete random variables

Definition. We define discrete random variables and the PMF.

- A *random variable* X is a function $S \to \mathbb{R}$ where S is a sample space. Note that the function $X : S \to \mathbb{R}$ is deterministic; for any fixed outcome $s \in S$, $X(s)$ is a single, well-defined number. The randomness in X comes solely through the randomness in the input outcome s.

- The *distribution* $\mathscr{L}_X$ of a random variable X is a function from the set of subsets of $\mathbb{R}$ to $[0, 1]$ specifying the probability X takes in values in each subset. That is, for $B \subseteq \mathbb{R}$, $\mathscr{L}_X(B) = P(X \in B)$, where $X \in B$ means $\{s \in S|X(s) \in B\}$. Note that random variables on different sample spaces can have the same distribution.

- A random variable X is *discrete* if there exists a finite or countably infinite set of values $a_1, a_2, \cdots$ such that $P(X = a_j \text{ for some } j) = 1$.

- Let X be a discrete random variable. The *probability mass function* (PMF) $p_X : \mathbb{R} \to [0, 1]$ with $p_X(x) = P(X = x)$. The distribution of a random variable specifies the PMF, and vice versa.

- The *support* of a discrete random variable X is the set of values that X can take on with positive probability. Formally, $\text{supp}(X) = \{x \in \mathbb{R}|p_X(x) > 0\}$.

Note. If $X = Y$, then $\mathscr{L}_X = \mathscr{L}_Y$, but the converse is not true.

Prop. If we know $\mathscr{L}_X$, then $p_X(x) = P(X = x) = P(X \in \{x\}) = \mathscr{L}_X(\{x\})$.

Note. If X is discrete,

$$\sum_{x \in \text{supp}(X)} p_X(x) = 1$$

Prop. Given a random variable X, its image $g(x)$ under a *transformation* $g : \mathbb{R} \to \mathbb{R}$ is also a random variable. For any transformation $g : \mathbb{R} \to \mathbb{R}$, $\mathscr{L}_X = \mathscr{L}_Y$ implies $\mathscr{L}_{g(X)} = \mathscr{L}_{g(Y)}$.

3 Named discrete distributions, expectation

An r.v. X is said to have the *Bernoulli distribution* with parameter p if $P(X = 1) = p$ and $P(X = 0) = 1 - p$, where $p \in (0, 1)$. We write $X \sim \text{Bern}(p)$; the symbol $\sim$ is read as "distributed as".

Let A be an event. Then the r.v. I_A with

$$I_A(s) = \begin{cases} 1 & \text{if } s \in A \\ 0 & \text{if } s \notin A \end{cases}$$

is called the *indicator random variable* for the event A. Then $I_A \sim \text{Bern}(P(A))$.

Suppose I flip a weighted coin n times and the coin lands heads with probability $p \in (0, 1)$. Then the r.v. X counting the number of heads flipped follows a *binomial distribution* with parameters n and p; we write $X \sim \text{Bin}(n, p)$.

Prop. If $X \sim \text{Bin}(n, p)$ its PMF is

$$p_X(x) = \binom{n}{x} p^x (1 - p)^{n - x}$$

where $x \in \text{supp}(X) = \{0, 1, \cdots, n\}$.

Suppose I flip a weighted coin with probability p of heads repeatedly. Let X be the number of tails I observe before flipping heads for the first time. Then X follows a *geometric distribution* with parameter p; we write $X \sim \text{Geom}(p)$. The distribution of $X + 1$ is $\text{FS}(p)$ (standing for first success).

Prop. If $X \sim \text{Geom}(p)$, its PMF is $p_X(x) = (1 - p)^x p$ where $x \in \text{supp}(X) = \{0, 1, \cdots\}$. If $Y \sim \text{FS}(p)$, then $P_y(Y) = P(X = y - 1) = (1 - p)^{y-1} p$ where $p \in \text{supp}(Y) = \{1, 2, \cdots\}$

A random variable X follows a *Poisson distribution* with parameter $\lambda > 0$ if $p_X(x) = \frac{\exp(-\lambda)\lambda^x}{x!}$, $x = 0, 1, \cdots$.

Theorem. (*Law of rare events*) Suppose events $A_1, \cdots, A_n$ occur with probabilities $p_1, \cdots, p_n$, respectively and are independent or nearly so. Then if

$$\sum_{i=1}^{n} p_i \to \lambda > 0$$

as $n \to \infty$, then the distribution of $\sum_{i=1}^{n} I_{A_i}$ is approximately $\text{Pois}(\lambda)$ for large n.

Let X be a discrete r.v. with PMF p_X. Then we define the *expectation value*

$$\mathbb{E}[X] = \sum_{x \in \text{supp}(X)} x p_X(x) = \sum_{x \in \text{supp}(X)} x P(X = x)$$

$\mathbb{E}[X]$ is determined by its PMF, and hence by its distribution.

Example. If $X \sim \text{Bern}(p)$, then $\mathbb{E}[X] = p$.

Prop. ("Fundamental bridge") $\mathbb{E}[I_A] = P(A)$

Example. If $X \sim \text{Pois}(\lambda)$,

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{x=0}^{\infty} x \frac{\exp(-\lambda)\lambda^x}{x!} \\
&= \sum_{x=1}^{\infty} \frac{\exp(-\lambda)\lambda^x}{(x-1)!} \\
&= \lambda \exp(-\lambda) \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \\
&= \lambda \exp(-\lambda) \sum_{x=0}^{\infty} \frac{\lambda^{x}}{(x-1)!} \\
&= \lambda
\end{aligned}$$

4 LotUS, linearity, variance

Theorem. (*Law of the unconscious statistician, or LotUS*) Let $g : \mathbb{R} \to \mathbb{R}$, and let X be a discrete r.v. with PMF p_X. Then

$$\mathbb{E}[g(X)] = \sum_{x \in \text{supp}(X)} g(x) p_X(x)$$

Example. Suppose $X \sim \text{Pois}(\lambda)$; to find $\mathbb{E}[e^{-X}]$,

$$\mathbb{E}[e^{-X}] = \sum_{x=0}^{\infty} e^{-x} \frac{e^{-\lambda}\lambda^x}{x!} = e^{\lambda(\frac{1}{e}-1)}$$

Theorem. (*Linearity*) Let X, Y be r.v.'s and a, b be constants. Then

(i) $\mathbb{E}[aX] = a\mathbb{E}[X]$

(ii) $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

Recall the fundamental brige; in cases when we can write $X = \sum_{i=1}^{n} I_{A_i}$, it might be useful to consider that $\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[I_{A_i}] = \sum_{i=1}^{n} P(A_i)$.

Prop. If $X \sim \text{Bin}(n, p)$, then $\mathbb{E}[X] = np$.
Proof. Begin with the coinflip story of $X \sim \text{Bin}(n, p)$, and let A_i be the event that the ith flip is heads, where $i = 1, \cdots, n$. We can write $X = \sum_{i=1}^{n} I_i$ (where $I_i \equiv I_{A_i}$); then, $\mathbb{E}[X] = \sum_{i=1}^{n} P(A_i) = np$.

Example. (2006 Putnam, A4) The numbers $1, \cdots, n$ are randomly ordered. A local minimum in the ordering is defined to be a number lower than all adjacent numbers in the ordering. What is the expected number of local minima?

To do this, let X be the number of local minima. We write $X = \sum_{i=1}^{n} I_{A_i}$, where A_i is the event that the ith number in the ordering is a local minimum. Then $\mathbb{E}[X] = \sum_{i=1}^{n} P(A_i)$. $P(A_1) = \frac{1}{2}$, since the probability of the number to the right of the first number being bigger than the first number is 1/2. By similar logic, $P(A_2) = \frac{1}{3}$, $\cdots$, $P(A_{n-1}) = \frac{1}{3}$, and $P(A_n) = \frac{1}{2}$. So the sum is $\mathbb{E}[X] = \frac{1}{2} + (n-2)\frac{1}{3} + \frac{1}{2} = \frac{n+1}{3}$.

Example. I draw cards without replacement one at a time. How many cards do I draw, on average, before getting my first ace?

Let X be the number of cards before the first ace, and let A_i be the event that the ith non-ace appears before all the aces. Then $X = \sum_{i=1}^{n} I_{A_i}$ and $\mathbb{E}[X] = \sum_{i=1}^{n} P(A_i)$. If you throw away all the other cards other than the ith non-ace and the four aces, the probability that the ith non-ace appears before all of the aces is $P(A_i) = \frac{1}{5}$, so $\mathbb{E}[X] = \frac{48}{5}$.

Theorem. If $X \sim \text{Geom}(p)$, then $\mathbb{E}[X] = \frac{1-p}{p}$.

Definition. The *variance* of a r.v. X is

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

and the *standard deviation* is $SD(X) = \sqrt{\text{Var}(X)}$.

Example. We give the variances for some distributions:

(i) $X \sim \text{Bern}(p) \implies \text{Var}(X) = p(1-p)$

(ii) $X \sim \text{Pois}(\lambda) \implies \text{Var}(X) = \lambda$

(iii) $X \sim \text{Geom}(p) \implies \text{Var}(X) = \frac{1-p}{p^2}$

Theorem. Suppose X is an r.v. Then for any constant c,

(i) $\text{Var}(cX) = c^2\text{Var}(X)$

(ii) $\text{Var}(X + c) = \text{Var}(X)$

Example. Let $X \sim \text{FS}(p)$. Since $X - 1 \sim \text{Geom}(p)$, we have $\text{Var}(X) = \text{Var}(X-1) = \frac{1-p}{p^2}$.

5 CDF, PDF, continuous random variables

Definition. The *cumulative distribution function (CDF)* of an r.v. X is a function $F_X : \mathbb{R} \to [0,1]$ with $F_X(x) = P(X \le x)$.

Prop. Here are some properties.

- The CDF is non-decreasing. $x < y \implies F_X(x) \le F_X(y)$. Since $\{X \le y\} \subset \{X \le y\}$, it follows from monotonicity that $P(X \le x) \le P(X \le y)$.
- $0 = \lim_{x\to-\infty} F_X(x) = 1 - \lim_{x\to\infty} F_X(x)$

- The CDF is right-continuous; $\lim_{x\to a^+} F_X(x) = F_X(a)$.

Example. Suppose $X \sim \text{Bern}(p)$. Then

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$$

Example. Suppose $X \sim \text{Geom}(p)$. Then

$$\begin{aligned} F_X(x) &= P(X \le x) \\ &= 1 - P(X > x) \\ &= 1 - (1-p)^{\lfloor x \rfloor + 1} \quad \text{for } x \ge 0 \end{aligned}$$

Definition. An r.v. X is *continuous* if its CDF F_X is continuous and differentiable at all but at most countably many points. For a continuous r.v. X, its *probability density function (PDF)* is

$$f_X(x) = \frac{d}{dx} F_X(x)$$

The support of a continuous r.v. X is

$$\text{Supp}X = \text{int}\{x | f_X(x) > 0\}$$

Prop. If X is a continuous r.v., then $P(X = a) = 0$ for all $a \in \mathbb{R}$.
Proof.

$$\begin{aligned} P(X = a) &\le P(a - h < X \le a) \quad \text{for all } h > 0 \\ &= P(X \le a) - P(X \le a - h) \\ &= F_X(a) - F_X(a-h) \end{aligned}$$

But in the limit $h \to 0^+$, by continuity $F_X(a) - F_X(a-h) \to 0$.

Prop. If f_X is the PDF of an r.v. then

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt$$

Note that $P(a \le X \le b) = P(a < X \le b) = P(a \le X < b) = P(a < X < b)$, because they differ from each other by the probability of $X = a$ or $X = b$, which is 0. In terms of the PDF,

$$P(a \le X \le b) = P(X \le b) - P(X \le a) = \int_a^b f_X(t)dt$$

Measure-theoretically,

$$\mathscr{L}_X(B) = P(X \in B) = \int_B f_X(t)dt$$

for any $B \subseteq \mathbb{R}$ for any reasonable B that you can get without trying really hard to get a misbehaved subset.

Of course, $\int_{-\infty}^{\infty} f_X(t)dt = 1$.

Definition. The *uniform distribution* with parameters $a < b$ has constant PDF on its support (a, b). We have

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

Note. If $X \sim \text{Unif}(a, b)$,

$$\begin{aligned} F_X(x) &= \int_a^x \frac{1}{b-a} dt \\ &= \frac{x-a}{b-a} \quad \text{for } a < x < b \end{aligned}$$

Definition. A r.v. X has the *exponential distribution* with parameter $\lambda > 0$ if its PDF is

$$f_X(x) = \lambda \exp(-\lambda x), \quad x > 0$$

Theorem. Suppose $X \sim \text{Expo}(\lambda)$. Then it satisfies *memorylessness*

$$P(X \geq t + s | X \geq s) = P(X \geq t)$$

for all $t, s \geq 0$.

Definition. If X is a continuous random variable,

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_{\text{Supp}(X)} x f_X(x) dx \end{aligned}$$

Theorem. If $g : \mathbb{R} \to \mathbb{R}$ is a function,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Example. Suppose $U \sim \text{Unif}(0, 1)$.

$$\begin{aligned} \mathbb{E}[U] &= \int_0^1 x dx = \frac{1}{2} \\ \text{Var}(U) = \mathbb{E}[U^2] - (E[U])^2 &= \frac{1}{12} \\ (\mathbb{E}[U^2] = \int_0^1 x^2 dx &= \frac{1}{3}) \end{aligned}$$

Example. Suppose $X \sim \text{Expo}(\lambda)$.

$$\begin{aligned} \mathbb{E}[X] &= \int_0^{\infty} x \lambda \exp(-\lambda x) dx \\ &= \frac{1}{\lambda} \\ \text{Var}(X) &= \frac{1}{\lambda^2} \end{aligned}$$

6 Normal distribution, univariate transformations

Definition. A random variable Z has the *normal distribution* with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$ if its PDF is

$$f_Z(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z-\mu)^2}{2\sigma^2}\right), \quad z \in \mathbb{R}$$

We write $Z \sim \mathcal{N}(\mu, \sigma^2)$. The *standard normal distribution* is $\mathcal{N}(0,1)$.

We denote by ϕ the standard normal PDF

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

and by Φ the standard normal CDF

$$\Phi(z) = \int_{-\infty}^{z} \phi(x)dx$$

Prop. For all $z \in \mathbb{R}$, $\phi(z) = \phi(-z)$ and $\Phi(z) = 1 - \Phi(-z)$.

Prop. Let X be an r.v. with CDF F_X. Then for any $\mu \in \mathbb{R}$, $\sigma > 0$, $Y = \mu + \sigma X$ has CDF $F_Y(y) = F_X\left(\frac{y-\mu}{\sigma}\right)$. If X is continuous, then Y is continuous with $f_Y(y) = \frac{1}{\sigma} f_X\left(\frac{y-\mu}{\sigma}\right)$

Prop. We can use the above theorem to show what happens to some distributions under some transformations of the r.v.

- If $Z \sim \mathcal{N}(0,1)$, then $Y = \mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$. It is useful to write $\frac{Y-\mu}{\sigma} \sim \mathcal{N}(0,1)$.
- Suppose $U \sim \mathrm{Unif}(0,1)$. Then, for any $a < b$, $X = (b-a)U + a \sim \mathrm{Unif}(a,b)$.
- Suppose $X \sim \mathrm{Expo}(\lambda)$. Then $Y = cX \sim \mathrm{Expo}(\lambda/c)$ for $c > 0$.

Prop. If $Z \sim \mathcal{N}(\mu, \sigma^2)$, then $\mathbb{E}[Z] = \mu$ and $\mathrm{Var}(Z) = \sigma^2$.

Theorem. Suppose X is a continuous r.v. and $g : \mathrm{supp}(X) \to \mathbb{R}$ is differentiable and either strictly increasing or decreasing. Then $Y = g(X)$ is continuous with PDF

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| \\ &= f_X(g^{-1}(y)) \left| g'(g^{-1}(y)) \right|^{-1} \end{aligned}$$

Proof. We show the strictly decreasing case; assume g is strictly decreasing. Then

$$\begin{aligned} F_Y(y) &= P(Y \le y) \\ &= P(g(X) \le y) \\ &= P(X \ge g^{-1}(y)) \\ &= 1 - P(X < g^{-1}(y)) \\ &= 1 - F_X(g^{-1}(y)) \\ f_Y(y) &= -f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) \\ &= f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| \end{aligned}$$

7 Joint and marginal distributions, independent r.v.'s

We move on to the section of the course that uses multiple random variables. For the sake of pretty notation, we only use two r.v.'s, but what follows can naturally be extended to more r.v.'s.

Definition. The *joint CDF* of two r.v.'s X and Y is the function $F_{XY} : \mathbb{R}^2 \to [0,1]$ with $F_{XY}(x,y) = P(X \leq x, Y \leq y)$. The *joint distribution* $\mathscr{L}_{XY} : \{\text{subsets of } \mathbb{R}^2\} \to [0,1]$ specifies $\mathscr{L}_{XY}(B) = P((X,Y) \in B)$ where $B \subseteq \mathbb{R}^2$ is any reasonable subset.

The joint CDF determines the joint distribution by

$$\mathscr{L}_{XY}((-\infty, x] \times (-\infty, y]) = P(X \leq x, Y \leq y) = F_{XY}(x,y)$$

Definition. If X and Y are discrete, their *joint PMF* $p_{XY} : \mathbb{R}^2 \to [0,1]$ specifies $p_{XY}(x,y) = P(X = x, Y = y)$. If X and Y are continuous, their *joint PDF* $f_{XY} : \mathbb{R}^2 \to [0, \infty)$ is $f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y)$. The *support* of (X,Y) is the interior of the set of all $(X,Y) \in \mathbb{R}^2$ where the joint PDF/PMF is strictly positive.

To get from the joint PMF to the joint CDF, we have

$$F_{XY}(x,y) = P(X \leq x, Y \leq y) = \sum_{x_0 \in \text{supp}(X) \cap (-\infty, x]} \sum_{y_0, \text{supp}(Y) \cap (-\infty, y]} P(X = x, Y = y)$$

In the continuous case, to get from the joint PDF to the joint CDF we simply have

$$F_{XY}(x,y) = P(X \leq x, Y \leq y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{XY}(a,b) db da$$

Of course,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x,y) dy dx = 1$$

Importantly,

$$\mathscr{L}_{XY}(B) = P((X,Y) \in B) = \iint_B f_{XY}(x,y) dy dx$$

Prop. If X, Y are discrete, we can obtain the *marginal distribution* (i.e. the distribution in one variable) from the joint distribution by

$$p_X(x) = \sum_{y \in \text{supp}(Y)} p_{XY}(x,y)$$

If X, Y are continuous,

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y) dy$$

Example. U is uniformily distributed on $B \subset \mathbb{R}^2$ if its support is B and it has a constant joint PDF on B. Of course,

$$f_{XY}(x,y) = \begin{cases} c & (x,y) \in B \\ 0 & (x,y) \notin B \end{cases}$$

with $c = \frac{1}{\iint_B dx dy} = \frac{1}{|B|}$ to satisfy normalization, and $P((X,Y) \in S) = \frac{|S|}{|B|}$.

Definition. R.v.'s $X_1, \cdots, X_n$ are *independent* if $P(\bigcap_{i=1}^n X_i \in B_i) = \prod_{i=1}^n P(X_i \in B_i)$ for all reasonable $B_1, \cdots, B_n \subset \mathbb{R}^2$.

Prop. $X_1, \cdots, X_n$ are independent iff

$$P(\bigcap_{i=1}^n X_i \leq x_i) = \prod_{i=1}^n P(X_i \leq x_i)$$

for all $x_1, \cdots, x_n \in \mathbb{R}$. Furthermore, if $X_1, \cdots, X_n$ are discrete, $X_1, \cdots, X_n$ are independent iff

$$P(\bigcap_{i=1}^n X_i = x_i) = \prod_{i=1}^n P(X_i = x_i)$$

and if $X_1, \cdots, X_n$ are continuous, they are independent iff

$$f_{X_1 \cdots X_n}(x_1, \cdots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

Note. Knowing the marginal distributions alone is not enough to specify a joint distribution. However, you can get a joint distribution from the marginal distributions of independent random variables–the joint PDF is the product of the marginal PDFs.

Example. $T_1 \sim \text{Expo}(\lambda_1)$ is independent of $T_2 \sim \text{Expo}(\lambda_2)$. Find $P(T_1 < T_2)$.

By independence, the joint PDF is given by the product

$$\begin{aligned} f_{T_1T_2}(t_1, t_2) &= f_{T_1}(t_1) f_{T_2}(t_2) \\ &= \lambda_1 \exp(-t_1\lambda_1)\lambda_2 \exp(-t_2\lambda_2) \\ P((T_1, T_2) \in B) &= \int_0^\infty \int_0^{t_2} \lambda_1 \exp(-t_1\lambda_1)\lambda_2 \exp(-t_2\lambda_2) dt_1 dt_2 \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \end{aligned}$$

Definition. R.v's $X_1, \cdots, X_n$ are *exchangeable* if $(X_1, \cdots, X_n)$ have the same distribution as $(X_{\pi(1)}, \cdots, X_{\pi(n)})$ for any permutation π on $\{1, \cdots, n\}$.

Note. Exchangeable random variables have the same marginal distribution.

Prop. If $X_1, \cdots, X_n$ are independent and identically distributed (i.i.d.), then they are exchangeable.

Proof. Let F be the common CDF. The CDF of $(X_{\pi(1)}, \cdots, X_{\pi(n)})$ is

$$P(X_{\pi(1)} \leq x_1, \cdots, X_{\pi(n)} \leq x_n) = \prod_{i=1}^n P(X_{\pi(i)} \leq x_i)$$

by independence for all π.

Example. Let (X, Y) be exchangeable, and $B = \{(x, y) | x < y\}$.

$$\begin{aligned} P(X < Y) &= P((X, Y) \in B) \\ &= \mathscr{L}_{XY}(B) \\ &= \mathscr{L}_{YX}(B) \\ &= P((Y, X) \in B) \\ &= P(X > Y) \end{aligned}$$

If they are continuous, $P(X - Y = 0) = 0$.

See the exercise published in the notes!!!

8 Conditional distributions, covariance

Theorem. (*Multivariate LotUS*) For any $g : \mathbb{R}^2 \to \mathbb{R}$,

$$\mathbb{E}[g(X,Y)] = \iint g(x,y) f_{XY}(x,y) dy dx$$

if X, Y are continuous, and

$$\mathbb{E}[g(X,Y)] = \sum_x \sum_y g(x,y) p_{XY}(x,y)$$

in the discrete case.

Definition. Suppose Y is a discrete r.v. If X is also discrete, the *conditional PMF of* X *given* Y is

$$p_{X|Y}(x|y) = \frac{P(X = x, Y = y)}{P(Y = y)} = P(X = x|Y = y)$$

If X is continuous, the *conditional PDF of* X *given* Y is

$$f_{X|Y}(x|y) = \frac{d}{dx} P(X \le x|Y = y)$$

Definition. Suppose Y is a continuous r.v. If X is discrete, the *conditional PMF of* X *given* Y is

$$p_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) p_X(x)}{f_Y(y)}$$

If X is continuous, the *conditional PMF of* X *given* Y is

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)}$$

Note. Suppose X and Y are independent. If X, Y are discrete,

$$\begin{aligned} p_{XY}(x) &= p_X(x) p_Y(y) \\ p_{X|Y}(x|y) &= p_X(x) \\ p_{Y|X}(y|x) &= p_Y(y) \end{aligned}$$

If X, Y are continuous,

$$\begin{aligned} f_{XY}(x,y) &= f_X(x) f_Y(y) \\ f_{X|Y}(x|y) &= f_X(x) \\ f_{Y|X}(y|x) &= f_Y(y) \end{aligned}$$

Example. Suppose a chicken lays $N \sim \text{Pois}(\lambda)$ eggs. Each egg hatches independently with probability p. Let X be the number of hatched eggs and Y the number of unhatched eggs. We find the joint PMF of (X, Y). To begin, note that $X + Y = N$ and $X|N = n \sim \text{Bin}(n, p)$.

$$\begin{aligned} p_{XY}(x,y) = P(X = x, Y = y) &= P(X = x, N = x + y) \\ &= P(X = x|N = x + y) P(N = x + y) \\ &= \left(\binom{x+y}{x} p^x (1-p)^y \right) \frac{\exp(-\lambda)\lambda^{x+y}}{(x+y)!} \\ &= \exp(-\lambda p) \frac{(\lambda p)^x}{x!} \exp(-\lambda(1-p)) \frac{(\lambda(1-p))^y}{y!} \end{aligned}$$

Motice that here we have $X \sim \text{Pois}(\lambda p)$ and $Y \sim \text{Pois}(\lambda(1-p))$.

Note. Bayes' rule and the law of total probability work as you expect with conditional PMFs/PDFs; if X, Y are continuous

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)}$$

and if X is discrete while Y is continuous,

$$p_X(x) = \int_{-\infty}^{\infty} p_{X|Y}(x|y)f_Y(y)dy$$

ADD THE X UNIF(0,1) Y RANDOM IN X FROM THE NOTES!!!

Definition. The *covariance* between r.v.'s X and Y is

$$\begin{aligned}\text{Cov}(X,Y) &= \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\end{aligned}$$

The *correlation* is

$$\text{Cor}(X,Y) = \frac{\text{Cov}(X,Y)}{SD(X)SD(Y)}$$

Note that $-1 \leq \text{Cor}(X,Y) \leq 1$.

Theorem. For any r.v.'s W, X, Y, Z and constant a, b, c, d,

1. $\text{Cov}(X,X) = \text{Var}(X)$
2. $\text{Cov}(X,a) = 0$
3. $\text{Cov}(X,Y) = \text{Cov}(Y,X)$
4. $\text{Cov}(aX + bY, cZ + dW) = ac\text{Cov}(X,Z) + ad\text{Cov}(X,W) + bc\text{Cov}(Y,Z) + bd\text{Cov}(Y,W)$ (*bilinearity*); obviously, the following sub-results follow:
   (i) $\text{Cov}(aX, Z) = a\text{Cov}(X,Z)$
   (ii) $\text{Cov}(X + Y, Z) = \text{Cov}(X,Z) + \text{Cov}(Y,Z)$

Theorem. If X is independent of Y, then $\text{Cov}(X,Y) = 0$ (and hence $\text{Cor}(X,Y) = 0$).

Note. The converse does not hold: uncorrelated r.v.'s may not be independent.

Example. Suppose $X \sim \mathcal{N}(0,1)$.

$$\text{Cov}(X, X^2) = E[X^3] - (E[X])(E[X^2])$$

FINISH THIS EXAMPLE

Prop. If $X_1, \cdots, X_n$ are uncorrelated (i.e. pairwise uncorrelated),

$$\text{Var}(X_1 + \cdots + X_n) = \sum_{i=1}^{n} \text{Var}(X_i)$$

Example. Suppose $X \sim \text{Bin}(n, p)$. Since X is distributed the same as the sum of a sum of indicators $I_1 + \cdots + I_n$, with $I_i \sim \text{Bern}(p)$. By independence of the indicators,

$$\begin{aligned}\text{Var}(X) &= \text{Var}(I_1 + \cdots I_n)\\ &= \text{Var}(I_1) + \cdots + \text{Var}(I_n)\\ &= np(1-p)\end{aligned}$$

Example. What is the variance of the number of aces in the top 5 cards of a standard deck?

Let X be the number of aces, and define the indicator I_i to be 1 if the ith card is an ace and 0 otherwise. So $X = \sum_i^5 I_i$.

$$\begin{aligned}\text{Var}(X) &= \text{Var}(I_1 + \cdots + I_5))\\ &= \text{Cov}(I_1 + \cdots + I_5, I + \cdots I_5)\\ &= \sum_{i=1}^{5} \text{Cov}(I_i, I_i) + \sum_{i \neq j} \text{Cov}(I_i, I_j)\\ \text{Cov}(I_i, I_i) &= \text{Var}(I_i) = \frac{1}{13} \cdot \frac{12}{13}\\ \text{Cov}(I_i, I_j) &= \mathbb{E}[I_i I_j] - \mathbb{E}[I_i]\mathbb{E}[I_j]\end{aligned}$$

We use $I_i \sim \text{Bern}(\frac{1}{13})$ and $I_i I_j \sim \text{Bern}\left(\frac{1}{13} \cdot \frac{3}{51}\right)$ to obtain the final solution.

9 Convolutions, multivariate transformations

Theorem. (*Convolution*) If X and Y are independent, let $T = X + Y$. If X, Y are discrete, then

$$p_T(t) = \sum_{x \in \text{supp}(X)} p_X(x) p_Y(t-x) = \sum_{y \in \text{supp}(Y)} p_Y(y) p_X(t-y)$$

If X, Y are continuous,

$$\begin{aligned}f_T(t) &= \int_{-\infty}^{\infty} f_X(x) f_Y(t-x) dx = \int_{-\infty}^{\infty} f_Y(y) f_X(t-y) dy\\ F_T(t) &= \int_{-\infty}^{\infty} f_X(x) F_Y(t-x) dx\end{aligned}$$

Theorem. $X \sim \text{Pois}(\gamma)$ independent of $Y \sim \text{Pois}(\mu)$ implies $T = X + Y \sim \text{Pois}(\lambda + \mu)$.

Proof. We demonstrate the use of the convolution formula to prove this fact:

$$\begin{aligned}p_T(t) &= \sum_{x=0}^{\infty} p_X(x) p_Y(t-x)\\ &= e^{-(\lambda+\mu)} (\lambda+\mu)^t \sum_{x=0}^{t} \frac{1}{x!(t-x)!} \left(\frac{\lambda}{\lambda+\mu}\right)^x \left(\frac{\mu}{\lambda+\mu}\right)^{t-x}\end{aligned}$$

Note that the PMF of $\text{Bin}\left(t, \frac{\lambda}{\lambda+\mu}\right)$ sums to 1:

$$\begin{aligned}1 &= \sum_{x=0}^{t} \binom{t}{x} \left(\frac{\lambda}{\lambda+\mu}\right)^x \left(1 - \frac{\lambda}{\lambda+\mu}\right)^{t-x}\\ &= t! \cdot S\end{aligned}$$

where $S = \sum_{x=0}^{t} \frac{1}{x!(t-x)!} \left(\frac{\lambda}{\lambda+\mu}\right)^x \left(1 - \frac{\lambda}{\lambda+\mu}\right)^{t-x}$, so $S = \frac{1}{t!}$, therefore $p_T(t) = e^{-(\lambda+\mu)} \cdot (\lambda+\mu)^t \cdot \frac{1}{t!}$.

Theorem. If $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ is independent of $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$. then $X + Y \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$.

Proof. First assume $\mu_X = \mu_Y = 0$.

$$\begin{aligned} f_T(t) &= \int_{-\infty}^{\infty} f_X(x) f_Y(t-x) dx \\ &\propto \exp\left(-\frac{t^2}{2\sigma_Y^2}\right) \exp\left(\frac{\sigma_X^2 t^2}{2(\sigma_X^2 + \sigma_Y^2)\sigma_Y^2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{\left(x - \frac{\sigma_X^2}{\sigma_X^2+\sigma_Y^2} t\right)^2}{2\frac{\sigma_X^2\sigma_Y^2}{\sigma_X^2+\sigma_Y^2}}\right) dx \\ &\propto \exp\left(-\frac{t^2}{2(\sigma_X^2 + \sigma_Y^2)}\right) \end{aligned}$$

where we have used the fact that the integral is $\sqrt{2\pi \frac{\sigma_X^2 \sigma_Y^2}{\sigma_X^2+\sigma_Y^2}}$. For general μ_X, μ_Y, write $X + Y = (X - \mu_X) + (Y - \mu_y) + \mu_X + \mu_Y$ and consider that $X - \mu_X \sim \mathcal{N}(0, \sigma_X^2)$ and $Y - \mu_Y \sim \mathcal{N}(0, \sigma_Y^2)$, so $(X - \mu_X) + (Y - \mu_Y) \sim \mathcal{N}(0, \sigma_X^2 + \sigma_Y^2)$ and we obtain our result.

Definition. The *Gamma function* is given by

$$\Gamma(x) = \int_0^{\infty} t^{x-1} \exp(-t) dt$$

This function has the property that $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$, and $\Gamma(a+1) = a\Gamma(a)$ for all $a > 0$.

Definition. A r.v. X has the *beta distribution* $X \sim \mathrm{Beta}(a, b)$ with $a, b > 0$ if its PDF is

$$f_X(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}$$

for $x \in (0, 1)$.

Note that $\mathrm{Beta}(1, 1) = \mathrm{Unif}(0, 1)$.

Definition. A r.v. Y has the *gamma distribution* $Y \sim \mathrm{Gamma}(a, \lambda)$ with $a, \lambda > 0$ if its pDF is

$$f_Y(y) = \frac{1}{y\Gamma(a)} (\lambda y)^a \exp(-\lambda y), \quad y > 0$$

Theorem. Suppose $X_1, \cdots, X_n$ are i.i.d. $\mathrm{Expo}(\lambda)$. Then $X_1 + \cdots X_n \sim \mathrm{Gamma}(n, \lambda)$.

Proof. We prove this for the case $n = 2$. Define $T = X_1 + X_2$.

$$\begin{aligned} f_T(t) &= \int_0^{\infty} f_X(x) f_Y(t-x) dx \\ &= \int_0^{t} \lambda \exp(-\lambda x) \lambda \exp(-\lambda(t-x)) dx \\ &= \lambda^2 \int_0^t \exp(-\lambda t) dx \\ &= t\lambda^2 \exp(-\lambda t) \end{aligned}$$

which is the PDF for $\mathrm{Gamma}(2, \lambda)$.

Theorem. Suppose $\mathbf{X} = (X_1, \cdots, X_n)$ is a vector of continuous r.v.'s with joint PDF $f_{\mathbf{X}}$ and suppose $g : \mathbb{R}^n \to \mathbb{R}^n$ is invertible on $\operatorname{supp}(\mathbf{X})$. Further assume that the Jacobian matrix of g^{-1} at $\vec{y}$

$$\frac{\partial \vec{x}}{\partial \vec{y}} = \begin{bmatrix} \frac{\partial x_1}{\partial y_1} & \cdots & \frac{\partial x_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \cdots & \frac{\partial x_n}{\partial y_n} \end{bmatrix}$$

exists for all $\vec{x} \in \operatorname{supp}(\mathbf{X})$, $\vec{y} = g(\vec{x})$. Then the joint PDF of $\mathbf{Y} = g(\mathbf{X})$ is

$$f_{\mathbf{Y}}(\vec{y}) = f_{\mathbf{X}}(\vec{x}) \left| \frac{\partial \vec{x}}{\partial \vec{y}} \right|$$

where $|\cdot|$ denotes the absolute value of the determinant.

Example. (Bank post-office story) Let $X \sim \text{Gamma}(a, \lambda)$ be the time that you waited at the bank and, independently, the $Y \sim \text{Gamma}(b, \lambda)$ the time you wait at the bank. We show that the total time spent waiting $T = X + Y \sim \text{Gamma}(a + b, \lambda)$ is independent of the fraction of time you wait at the post office, $W = \frac{X}{X+Y} \sim \text{Beta}(a, b)$.

Let $(T, W) = g(X, Y)$ where we regard (T, W) and (X, Y) as vectors $\vec{y}$ and $\vec{x}$ respectively. We have $g(x, y) = (x + y, \frac{x}{x+y})$, so g^{-1} has

$$x + y = t, \quad x = tw, \quad \frac{x}{x+y} = w, \quad y = t(1 - w)$$

Then

$$\begin{aligned} \frac{\partial(x, y)}{\partial(t, w)} &= \begin{pmatrix} w & t \\ 1 - w & -t \end{pmatrix} \\ \left| \frac{\partial(x, y)}{\partial(t, w)} \right| &= |-wt - t(1 - w)| = t \\ f_{(T,W)}(t, w) &= f_{(X,Y)}(x, y)t \\ &= f_X(x) f_Y(y) t \end{aligned}$$

Plugging things in and doing algebra gives

$$f_{(T,w)}(t, w) = \left(\frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} w^{a-1}(1 - w)^{b-1} \right) \left(\frac{1}{\Gamma(a + b)} (\lambda t)^{a+b} e^{-\lambda t} \frac{1}{t} \right)$$

for $w \in (0, 1)$ and $t > 0$. Since the joint PDF factors into a function of t times a function of w, we have that T and W are independent. We recognize the marginal PDFs of T and W and deduce $T \sim \text{Gamma}(a + b, \lambda)$ and $W \sim \text{Beta}(a, b)$.

10 Conditional expectation

Definition. Suppose A is an event with $P(A) > 0$ and X is a r.v. Then the *conditional distribution* of X given A is $\mathscr{L}_{X|A}(B) = P(X \in B|A)$ and the *conditional expectation* of X given A is the expectation of a r.v. with distribution $\mathscr{L}_{X|A}$. In the discrete case,

$$\mathbb{E}[X|A] = \sum_{x \in \operatorname{supp}(X)} x P(X = x|A)$$

and in the continuous case,

$$\mathbb{E}[X|A] = \int_{-\infty}^{\infty} x f_{X|A}(x)dx$$

Prop. For any event A with $P(A) > 0$, $\mathbb{E}[X|A] = \frac{\mathbb{E}[XI_A]}{P(A)}$.

Theorem. (*Law of total expectation*) Suppose X is a r.v. and $B_1, \cdots, B_n$ partition the sample space with $P(B_1 > 0)$ for all i. Then

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X|B_i]P(B_i)$$

Definition. Suppose X, Y are r.v.'s. If Y is discrete, let $\mathbb{E}[X|Y = y]$ for all $y \in \text{supp}(Y)$; if Y is continuous, let $g(y)$ be the expectation specified by the conditional distribution of X given $Y = y$. Then $\mathbb{E}[X|Y] = g(Y)$.

Prop. Let X, Y, Z be r.v.'s.

- (*Takeout*) $\mathbb{E}[h(X)Y|X] = h(X)\mathbb{E}[Y|X]$
- If X, Y are independent, then $\mathbb{E}[Y|X] = \mathbb{E}[Y]$
- $\mathbb{E}[aX + bY|Z] = a\mathbb{E}[X|Z] + b\mathbb{E}[Y|Z]$

Theorem. (*Law of iterated expectations*) For any r.v.'s X, Y we have $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]]$.

11 Conditional variance, inequalities

Definition. The *conditional variance* of X given Y is

$$\begin{aligned} \text{Var}(X|Y) &= \mathbb{E}[(X - \mathbb{E}[X|Y])^2|Y] \\ &= \mathbb{E}[X^2|Y] - (\mathbb{E}[X|Y])^2 \end{aligned}$$

Theorem. (*Law of total variance*) For any r.v.'s X, Y,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X|Y)] + \text{Var}(\mathbb{E}[X|Y])$$

Theorem. (*Markov's inequality*) Let X be a r.v. with $\mathbb{E}(|X|) < \infty$. Then for any $a > 0$,

$$P(|X| \geq a) \leq \frac{\mathbb{E}[|X|]}{a}$$

Theorem. (*Chebyshev's theorem*) Suppose X has $\mathbb{E}[|X|] < \infty$ and $\text{Var}(X) > 0$. Then for $a > 0$,

$$P(|X - \mathbb{E}[X]| \geq aSD(X)) \leq \frac{1}{a^2}$$

Theorem. (*Chernoff bound*) For any r.v. X, and constants $t > 0$ and $a > 0$, we have

$$P(X \geq a) \leq \frac{\mathbb{E}[\exp(tX)]}{\exp(ta)}$$

Theorem. (*Cauchy-Schwartz inequality*) Suppose X and Y are r.v.'s with $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$. Then

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}$$

Definition. A function $g : I \to \mathbb{R}$ is *convex* on the interval $I \subseteq \mathbb{R}$ if

$$g(\lambda x + (1-\lambda)y) \leq \lambda g(x) + (1-\lambda)g(y)$$

for all $0 \leq \lambda \leq 1$, $x, y \in I$.

Theorem. (*Jensen's inequality*) Suppose X is an r.v. and g is convec on $\text{supp}(X)$. Then $\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$.

12 Law of large numbers, central limit theorem

Definition. Given r.v.'s $X_1, \cdots, X_n$, their *sample mean* is

$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n}$$

Prop. Suppose $X_1, \cdots, X_n$ are i.i.d. from some distribution mean μ, variance $\sigma^2 < \infty$. Then $\mathbb{E}[\overline{X}_n] = \mu$ and $\text{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$.

Theorem. (*Weak law of large numbers*) If $X_1, \cdots, X_n$ are i.i.d. with mean μ, variance $\sigma^2 < \infty$, then $\overline{X}_n \xrightarrow{p} \mu$ as $n \to \infty$; i.e., for any $\epsilon > 0$,

$$\lim_{n\to\infty} P(|\overline{X}_n - \mu| > \epsilon) = 0$$

Note. We assumed $\sigma^2 < \infty$; note that, from this condition follows $\mu < \infty$.

Theorem. (*Strong law of large numbers*) If $X_1, \cdots, X_n$ are i.i.d. with mean $\mu < \infty$, then the event that $\overline{X}_n$ converges to μ has probability 1, i.e. $X_n \xrightarrow{a.s.} \mu$.

Example. (*Empirical CDF*) Suppose $X_1, \cdots, X_n$ are i.i.d. and define the empirical CDF

$$\hat{F}_n(t) = \frac{1}{n}\sum_{i=1}^{n} I_{X_i \leq t} = \overline{Y}_n$$

By the law of large numbers,

$$\overline{Y}_n = \hat{F}_n(t) \xrightarrow{p} \mathbb{E}[Y_i] = \mathbb{E}[I_{X_i \leq t}] = P(X_i \leq t) = F(t)$$

i.e. the empirical CDF converges pointwise for every t to the true CDF.

Example. (Monte Carlo integration) Suppose $f : [a,b] \to [0,\infty)$ is continuous. This implies there exists $C < \infty$ s.t. $f(x) \leq c$ for all $a \leq x \leq b$. Suppose $(X_1, Y_1), \cdots, (X_n, Y_n)$ are i.i.d. points chosen independently and uniformly in the rectangle $[a,b] \times [0,c]$. Define

$$\overline{p}_n = \frac{1}{n}\sum_{i=1}^{n} I_{Y_i \leq f(X_i)}$$

Then,

$$\overline{p}_n \xrightarrow{p} \mathbb{E}[I_{Y_i \leq f(X_i)}] = P(Y_i \leq f(X_i)) = \frac{\int_a^b f(x)dx}{c(b-a)}$$

Theorem. (*Central limit theorem*) Suppose $X_1, \cdots, X_n$ are i.i.d. with mean μ, variance $\sigma^2 < \infty$. Let

$$S = \sqrt{n}\left(\frac{\overline{X}_n - \mu}{\sigma}\right)$$

Then $S_n \xrightarrow{d} \mathcal{N}(0,1)$, meaning $F_{S_n}(t) \to \Phi(t)$ for all t.

Example. If you flip 2000 fair coins, what's the probability more than 51% land heads? Let W be the number of head flips. Because $W \sim \text{Bin}\left(n, \frac{1}{2}\right)$, as $n \to \infty$.

$$\begin{aligned} P(W > 0.51n) &= \sum_{w=0.51n+1}^{2000} \binom{2000}{w} 0.5^w 0.5^{2000w} \\ &= \sum_{w=1021}^{2000} \binom{2000}{w} 0.5^w 0.5^{2000w} \end{aligned}$$

$W = I_1 + \cdots + I_{2000}$ where I_i is the indicator r.v. that the ith flip is heads; $I_i \sim \text{Bern}\left(\frac{1}{2}\right)$, so $\mathbb{E}(I_i) = \frac{1}{2}$, $SD(I_i) = \sqrt{\frac{1}{2}\left(1 - \frac{1}{2}\right)} = \frac{1}{2}$. We have

$$\begin{aligned} S_n &= \sqrt{n}\left(\frac{\overline{I}_n - \frac{1}{2}}{\frac{1}{2}}\right) \\ P(\overline{I}_n > 0.51) &= P\left(S_n > \sqrt{n}\left(\frac{0.51 - 0.5}{0.5}\right)\right) \\ &\longrightarrow 1 - \Phi\left(\frac{2\sqrt{5}}{5}\right) \end{aligned}$$

as $n \to \infty$.

Note that

$$S_n = \sqrt{n}\left(\frac{\overline{X}_n - \mu}{\sigma}\right) \approx \mathcal{N}(0,1)$$

so $\overline{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$. The *continuity correction* is as follows. For $X \sim \text{Bin}(n,p)$, write $P(X = k) \approx P\left(k - \frac{1}{2} < Y < k + \frac{1}{2}\right)$ for $Y \sim \mathcal{N}(np, np(1-p))$. For the coin flipping example, $P(\overline{I}_n > 0.51) = P(W > 1020) \approx P(1020.5 < Y < 2000.5)$. This yields a *much better* approximation than our initial guess from the example; while the first is off by almost 1%, the continuity correction matches the result up to 0.001%!

Example. Suppose $X \sim \text{Pois}(n)$. For large n, $X \sim \mathcal{N}(n,n)$. This is because $X \stackrel{d}{=} X_1 + \cdots + x_n$ where $X_i \stackrel{i.i.d.}{\sim} \text{Pois}(1)$.

Example. Suppose $X \sim \text{Gamma}(n, \lambda)$. For large n, $X \sim \mathcal{N}\left(\frac{n}{\lambda}, \frac{n}{\lambda^2}\right)$ because $X \stackrel{d}{=} X_1 + \cdots + x_n$ for $X_i \stackrel{i.i.d}{\sim} \text{Expo}(\lambda)$.

13 Moment generating functions

Definition. Let X be a r.v. Its *moment generating function MGF* is $M_X(t) = \mathbb{E}[\exp(tX)]$. The MGF "exists" as long as it is finite for all t in some open interval containing 0.

Prop. Suppose X has an MGF M_X. Then the MGF of $Y = a + bX$ is $M_Y(t) = \exp(ta)M_X(bt)$.

Prop. For $X_1, \cdots, X_n$ independent and $Y = X_1 + \cdot + X_n$, then for any t where $M_{X_i}(t) < \infty$ for all i, $M_Y(t) = \prod_{i=1}^n M_{X_i}(t)$.

Theorem. The MGF determines the distribution; more precisely, if $M_X(t) = M_Y(t)$ for all $-a < t < a$ for some $a > 0$, then $\mathscr{L}_X = \mathscr{L}_Y$.

The nth *moment* of an r.v. is $\mathbb{E}[X^n]$.

Prop. If M_X exists, the nth moment $\mathbb{E}[X^n] = M_X^{(n)}(0)$. Equivalently, $\mathbb{E}[X^n]$ is the coefficient of $\frac{t^n}{n!}$ in the Taylor series expansion for $M_X(t)$ about $t = 0$.

Theorem. (*Levy's continuity theorem*) Suppose M is the MGF of a continuous r.v. X and $\lim_{n\to\infty} M_n(t) = M(t)$ for all $t \in (-a, a)$ where M_n is the MGF of X_n. Then the CDFs $F_1, F_2, \cdots$ of $X_1, X_2, \cdots$ converge pointwise to F, the CDF of X.

The last thing we'll do in this class is prove the central limit theorem for a special case (with a relatively weak constraint). Let $X_1, X_2, \cdots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$, $\mathrm{Var}(X_i) = \sigma^2 < \infty$, and let $\tilde{X} = \frac{X_i - \mu}{\sigma}$, so the $\tilde{X}_i$ are i.i.d. with $\mathbb{E}[X_i] = 0$, $\mathrm{Var}(\tilde{X}_i) = 1$. Let M be the common MGF of the X_i (which we assume exists for this proof to work):

$$\tilde{M}(t) = \exp\left(-\frac{\mu}{\sigma}t\right) M\left(\frac{t}{\sigma}\right)$$

Let

$$Y_n = \sqrt{n}\overline{\tilde{X}}_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \tilde{X}_i\right)$$

Then

$$\begin{aligned}
M_{Y_n}(t) &= \mathbb{E}\left[\exp\left(\frac{t}{\sqrt{n}}\sum_{i=1}^n \tilde{X}_i\right)\right] = \prod_{i=1}^n \mathbb{E}\left[\exp\left(\frac{t}{\sqrt{n}}\tilde{X}_i\right)\right] = \left[\tilde{M}\left(\frac{t}{\sqrt{n}}\right)\right]^n \\
\tilde{M}\left(\frac{t}{\sqrt{n}}\right) &= \tilde{M}(0) + \tilde{M}'(0) + \frac{\tilde{M}''(0)}{2}\frac{t^2}{n} + o(n^{-1}) \\
&= 1 + \frac{t^2}{2n} + o(n^{-1}) \\
\log M_{Y_n}(t) &= n \log \tilde{M}\left(\frac{t}{\sqrt{n}}\right) \\
&= n \log\left(1 + \frac{t^2}{2n} + o(n^{-1})\right) \\
&= \frac{t^2}{2} + o(1)
\end{aligned}$$

$$\implies \log M_{Y_n}(t) \to \exp\left(\frac{t^2}{2}\right)$$